

6	(a)	(i)	PSYV <u>and</u> QRXW	B1
		(ii)	electrons moving in magnetic field deflected towards face QRXW / electrons accumulate on face QRXW	M1
			So face PSYV is more positive	A1
	(b)	(i)	Arrow point up the page	B1
		(ii)	$Eq = Bqv$ $v = \frac{E}{B} = \frac{12 \times 10^3}{930 \times 10^{-6}}$	C1
			$= 1.3 \times 10^7 \text{ m s}^{-1}$	A1
		(iii)	$Bqv = m v^2 / r$ $q/m = (1.3 \times 10^7) / (7.9 \times 10^{-2} \times 930 \times 10^{-6})$	C1
			$= 1.8 \times 10^{11} \text{ C kg}^{-1}$	A1

7	(a)	(i)		A1
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